

EDUCATION	Massachusetts Institute of Technology	MA, US
	<i>Ph.D. candidate in Physics</i> • Advisor: Prof. Sarah C. Millholland • Research area: Exoplanet dynamics and demographics • GPA: 5.0/5.0	2023 - Present
	Indian Institute of Science Education and Research Berhampur	Odisha, India
	<i>Integrated Bachelor's and Master's, BS-MS (Dual degree)</i> • Major: <i>Physical Sciences</i>; Minor: <i>Chemical Sciences</i> • Master's Thesis: <i>Study of Tidal Synchronization and Spots in Low Mass Eclipsing Binaries with TESS</i> • GPA: 9.6/10.0 (<i>Gold Medalist</i>)	2018 - 2023
FELLOWSHIPS AND AWARDS	• Quad Fellow, one of 37 students selected among 11 countries (\$20,000) • Whiteman Fellow, MIT Physics Department fellowship (\$109,000) • Director's Gold Medal for outstanding all round achievement and leadership in the graduating class across all disciplines of the BS-MS program at IISER Berhampur • Dr. K.K. & Dr. K.R. Chary Gold Medal for excelling in academics and sports in the graduating class across all disciplines of the BS-MS program at IISER Berhampur • DST INSPIRE Fellow, awarded to top 1% students in India (~\$3,150) • Silver Medal in International Natural Sciences Tournament • Bronze Medal in International Natural Sciences Tournament • Gold Medal in International Genetically Engineered Machine (iGEM), also nominated for the Best Diagnostic project	2025-2026 2023 - 2024 2023 2023 2018-2023 2022 2021 2021
PUBLICATIONS	First author publications	
	1. Ritika Sethi, Sarah Millholland & Tim Hallatt "Coupled Orbital and Interior Evolution of Sub-Neptunes" <i>ApJ</i>, 2026 2. Ritika Sethi, Morgan MacLeod & Sarah Millholland "Hydrodynamic Atmospheric Escape Rates from Planets in the Stellar Tidal Field" <i>ApJ</i>, 2026 3. Ritika Sethi, David V. Martin et al. "EBLM XVII- Tidal Synchronization and Circularization in Tight Stellar Binaries" <i>MNRAS</i>, 2026 4. Ritika Sethi & Sarah Millholland. "Tidal Inflation is Stronger for Misaligned Neptune-Sized Planets Than Aligned Ones" <i>ApJ</i>, 2025 5. Ritika Sethi & David V. Martin "Tight stellar binaries favour active longitudes at sub-and anti-stellar points." <i>MNRAS</i>, 2024 6. Ritika Sethi, D. Bisht et al. "A deep study of an Open Cluster NGC 5288 using Photometric and Astrometric data from Gaia DR3" <i>RMxAA</i>, 2023	
	Co-author Publications	
	1. Saugata Barat, Jean-Michel Désert, Allona Vazan, Ritika Sethi et al. "Investigating potential causes for high inferred internal temperatures of warm transiting planets" <i>AAS Journals</i> (in prep.) 2. Becca Spejcher, David V. Martin, Jake Pandina, Andy Zhang, Max Ammons, Wata Tubthong, Amaury Triaud, Ritika Sethi et al. "The EBLM Project XVIII. 3D Obliquities of Five Low-Mass Eclipsing Binaries" <i>MNRAS</i>, 2026	

3. Becca Spejcher, David V. Martin, Jake Pandina, Andy Zhang, Max Ammons, Wata Tubthong, Amaury Triaud, **Ritika Sethi** et al. “EBLM XVI- Spin-orbit alignment of the low mass eclipsing binary EBLM J0021-16” *MNRAS*, 2026
4. Sydney Jenkins, Andrew Vanderburg, **Ritika Sethi**, et. al. “An Eccentric Sub-Neptune Moving Into the Evaporation Desert” *ApJL*, 2025
5. David V Martin, Tayt Armitage, Alison Duck, Matthew I Swayne, Romy Rodríguez Martínez, **Ritika Sethi**, et al. “Revised temperatures for two benchmark M-dwarfs – outliers no more” *MNRAS*, 2024
6. David V Martin, **Ritika Sethi**, et al. “The benchmark M dwarf eclipsing binary CM Draconis with TESS: spots, flares, and ultra-precise parameters” *MNRAS*, 2024
7. Alison Duck et al. including **Ritika Sethi** [26 Total] “The EBLM project X. Benchmark masses, radii, and temperatures for two fully convective M-dwarfs using K2” *MNRAS*, 2023

GRANTS

EBLM M-dwarf Fundamental Parameters, TESS Cycle 7 (GO7005): Co-I (PI: David Martin)-\$70,000 2024

TALKS & POSTERS (* = INVITED)

Talk: “Coupled Orbital and Interior Evolution of Sub-Neptunes”, Emerging Researchers in Exoplanet Science (ERES) Symposium XI, University of Michigan, Ann Arbor 2026

Talk: “Coupled Orbital and Interior Evolution of Sub-Neptunes”, CT Exoplanet Meeting, Wesleyan University, Middletown, Connecticut 2026

Talk: “Modeling the Age-Dependent Occurrence Rates of Close-in Sub-Neptunes”, Division of Dynamical Astronomy (DDA) Meeting 2026

Talk: “An automated framework for evaluating alignment of scientific research with real-world policy decisions”, Quad Spring Symposium (Virtual) 2026

*Talk: “Hydrodynamic Atmospheric Escape Rates from Planets in the Stellar Tidal Field”, Journal Club, MIT, Cambridge, MA 2026

Poster: “Modeling the Age-dependent Occurrence Rates of Close-in Sub-Neptunes”, AASTCS 11: Exoplanet Atmospheres 2026, Denver, CO 2026

Talk: “Misaligned Neptune-Sized Planets are Puffier than the Aligned Ones”, Exoplanet Pizza Lunch, Harvard University, Cambridge, MA 2025

Talk: “Misaligned Neptune-Sized Planets show Stronger Tidal Inflation than the Aligned Ones”, CT Exoplanet Meeting, Wesleyan University, Middletown, Connecticut 2025

Talk: “Misaligned Neptunes are More Tidally Inflated than Aligned Ones”, ERES Symposium X, Princeton University, New Jersey 2025

Poster: “Misaligned Neptunes are more Tidally Inflated”, Division on Dynamical Astronomy (DDA) Meeting, Georgia Institute of technology, Atlanta 2025

Poster: “Exploring the Origins of the Polar Planets”, Exoplanet V conference, Leiden, Netherlands 2024

*Talk: “Tight stellar binaries favor active longitudes at sub-and anti-stellar points”, University of Wisconsin–Madison 2023

*Talk: “Tidal Synchronization and Stellar Activity in Tight Eclipsing Binaries”, Tufts University 2023

Talk: “Tidal synchronization and Spot Distribution in Low Mass Eclipsing Binaries”, IISER Berhampur, India 2022

SCIENCE COMMUNICATION

Planetarium Show Designer: Designing a planetarium show for middle-school students through MIT Kavli Portable Planetarium 2026-Present

MIT Sidewalks AstroGazers | Member: Organize outreach activities for general public in the Cambridge, and Boston area. 2023 - Present

Talk: Astronomy Club, International School of Boston 2025
Talk: Rashtriya Raksha University, Gujarat, India 2024
Article: “Podcasts as tools for science communication” for MIT Student Policy Review (SPR), 2024
Workshop: Lexington Middle School, MA 2024
Talk: “Graduate School Applications”, IISER Berhampur, Odisha, Indi 2023
Special Olympics, Volunteer: Organized events to promote inclusivity of specially-abled individuals. 2016-2023

LEADERSHIP & SERVICE

MIT Physics Graduate Student Council: Chair 2026 - Present
Tang Hall (MIT Graduate Housing) Residents Association: Chair 2025 - Present
LOC Member: Rogue Worlds 3 Conference 2026
MIT Graduate Students Activities Council: Member 2023 - 2024
Team Lead: Team India in the International Natural Sciences Tournament 2022
Secretary: Student Affairs Council (SAC) at IISER Berhampur 2018-2022
Co-Leader of Team IISER Berhampur for International Genetically Engineered Machine (iGEM) Competition 2021
Fest Coordinator of Tenacity-All IISER Virtual Games and Sports Fest comprising 10 universities across India 2021

MENTORSHIP & ADVOCACY

MIT Graduate & Undergraduate Womxn in Physics | Mentor 2024 - Present
Grads Advising Grad Admissions | Mentor 2025

TEACHING EXPERIENCE

18.06: Linear Algebra, MIT | Grader 2025
8.03: Physics III: Vibrations and Waves, MIT | Grader 2025
8.01: Introduction to Classical Mechanics, MIT | Grader 2024
8.02: Introduction to Electromagnetism and Electrostatics, MIT | Grader 2023
PHY201: Electromagnetism, IISER Berhampur | Teaching Assistant 2022

SELECTED MEDIA & FEATURES

MIT Physics: *Physics graduate student, Ritika Sethi, named 2025 Quad Fellow*
Apeejay Newsroom: *“Success is not a destination, it’s like chasing the horizon”*
IISER Berhampur: *Major Student and Alumni Achievements*
Quad Fellowship: *2025 Quad Fellows*
IISER Berhampur: *Some of Our Achievers*
Apeejay Newsroom: *MIT scholar advises study abroad aspirants*
News Features: *The Hans India*, and *Odisha Bytes*.